

Rwanda

Centre for the
Fourth Industrial
Revolution

AGRICULTURE USE CASE • INFO SESSION

AI-Enabled Agricultural Advisory Solution Partner

What we are building, how the benchmark works, and what we need from you

Rwanda AI Scaling Hub • 1 October 2026



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A sovereign, Kinyarwanda-first advisory platform

C4IR is seeking a Solution Partner to help design and build a multi-channel AI advisory platform.



Crop and livestock advisory

Planting, weeding and harvest schedules; animal health; soil and input guidance.



Pest and disease management

Identifying pests and diseases, forecasting risk, advising on treatment.



Weather and climate intelligence

Planting windows, drought and heavy-rain alerts, climate-smart practices.



Input and market access

Where to get inputs such as seed and fertilizer, and where to sell produce.

REACHING FARMERS WHERE THEY ARE

IVR (voice)

USSD / SMS

WhatsApp

Mobile / web app

On any channel, shaped around farmers' lived realities: internet connectivity, the device they own and their literacy level. Kinyarwanda first.

WHAT C4IR PREFERS

Conversational AI in Kinyarwanda

LLM-based agents, with strong voice

Offline-tolerant design

Usable on constrained rural networks

Multi-channel delivery

With image-based pest identification

Compliant with Rwanda's data law

Open source preferred, not a deal-breaker

We are building one sovereign, Kinyarwanda-first platform across four core use cases, reaching farmers on any channel that fits their connectivity, device and literacy, compliant with Rwanda's data laws.

The right advice isn't reaching farmers on time

Agriculture is 20% of GDP, over a third of exports and the livelihood of nearly 70% of households, and PSTA 5 targets 6.1% sector growth by 2028/29. However, all of this growth depends on farmers getting the right information, and today too few of them do.

THE REACH GAP

35%

of 3.4 million farmers reached by
16,654 extension agents

3 in 10

calls to the agricultural call centre
answered

WHERE EXTENSION REACHES, IT WORKS

+38%

yields among farmers served by
Twigire Muhinzi

+75%

milk production among the farmers
it serves

However, even where it reaches, the advice still falls short:



Generic

Not tailored to a farmer's soil, crop
or region.



Not personalised

Too few experts to fit answers to
each farm.



No feedback loop

Farmers' real problems don't reach research and policy
in real time.

Extension works where it reaches, but too few farmers get timely, tailored advice. The platform must reach everyone, fit each farm, and feed what farmers need back to MINAGRI and RAB.

Three user groups, three sets of needs

Farmers come first. The platform also has to work for the people who support them, and for those who set policy.

PRIMARY

Smallholder farmers

Median age 42, near-universal phone ownership, many on feature phones; most farm under 0.5 ha.

WHAT THEY NEED

- ✓ Direct, personalised answers
- ✓ On crops and livestock, pests and disease, weather
- ✓ By voice and simple text (SMS) preferably

SECONDARY

Extension officers, field agents, farmer promoters

Not replaced: made more efficient and consistent.

WHAT THEY NEED

- ✓ Digital record-keeping
- ✓ An in-field knowledge refresher
- ✓ A second opinion, e.g. a photo of a sick plant
- ✓ A way to escalate recurring issues

TERTIARY

MINAGRI, RAB and other agencies

No farmer-facing advice: they need visibility.

WHAT THEY NEED

- ✓ What farmers are asking
- ✓ Where the gaps are
- ✓ How needs trend, in real time rather than periodic surveys

Smallholder farmers are to be served first, however the extension ecosystem and government agencies that support them are to also be served by the platform. Their needs will be clarified further through user research as we build the platform.

A standardised test for every solution

What is a benchmark? A fixed, standardised test: the same items, the same conditions and the same scoring for every system, so results are fair, comparable and reproducible.



Text-based

Clean text questions in English and Kinyarwanda: the main event.



Your model and its harness

Prompts, retrieval, guardrails and orchestration: the whole brain.



Voice, optional

Kinyarwanda speech in and out: a bonus, and a tie-breaker when needed.



Two runs

Once with no local context, then again after you add it.



Real farmer questions

Genuine questions from Rwandan farmers, validated by agronomists.



A golden benchmark dataset

Kept blind: your solution sees the questions, but you never see the answers.

Both runs of the evaluation use the same golden benchmark dataset, under the same conditions. The difference between the two benchmarks will show the companies ability to improve on local data.

Two layers: text decides, Layer 2 adds value

Layer 1 is required and decides who advances. Layer 2 is added value, used mostly to break ties.

★ REQUIRED • DECIDES WHO ADVANCES

Layer 1 • Text

Your model and harness answer real farmer questions as clean text.

- ✓ English and Kinyarwanda, the same questions in both
- ✓ Every answer scored against our rubric by expert agronomists
- ✓ Run twice: baseline and final evaluation after solution augmentation

ADDED VALUE • TIE-BREAKER

Layer 2

Looked at only after the text results, mostly to break ties between close scores.



Voice

Kinyarwanda speech-to-text and text-to-speech, if you offer them.



No voice yet? Not a disqualifier



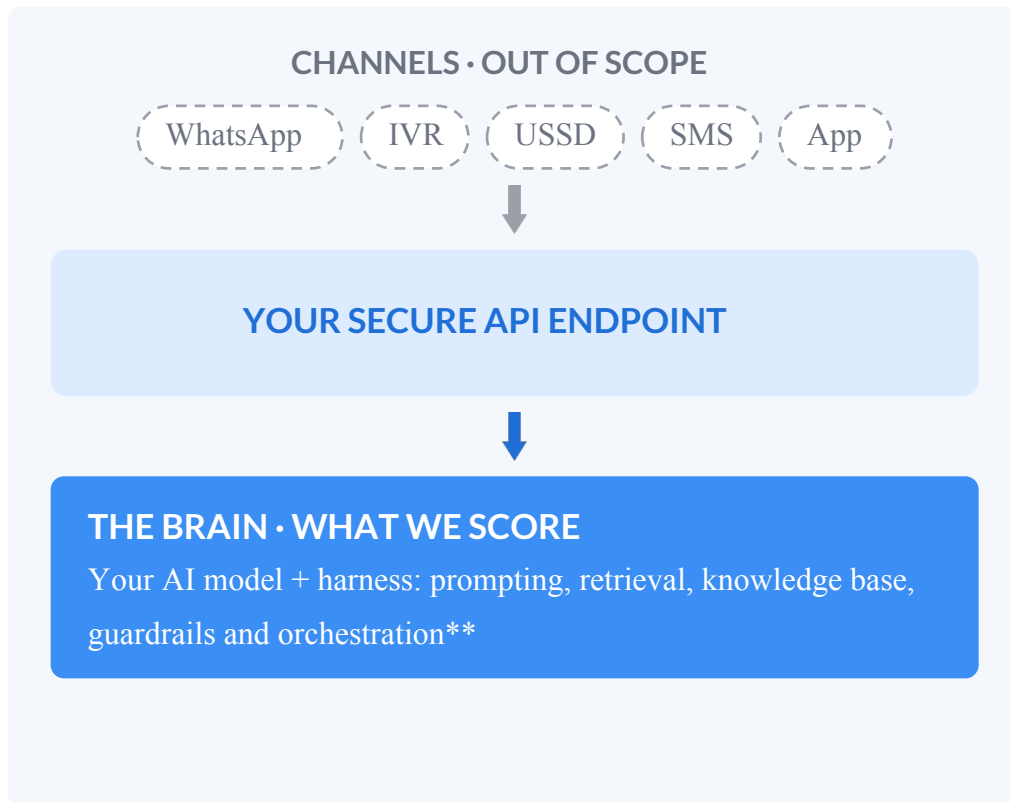
Lightweight A/B testing

Quick tests with small groups of real users on your existing channel.

Text decides who advances. Layer 2 adds value on top: we use it mostly to break ties between close text results.

We benchmark intelligence, over text

Wrap your model, its harness and everything around it behind one secure API.



The intelligence is what's certain

We are building this product together. Channels can change as we go; the brain is what we are choosing now.



English and Kinyarwanda, side by side

The same questions in both languages show how much a solution gives up in Kinyarwanda.



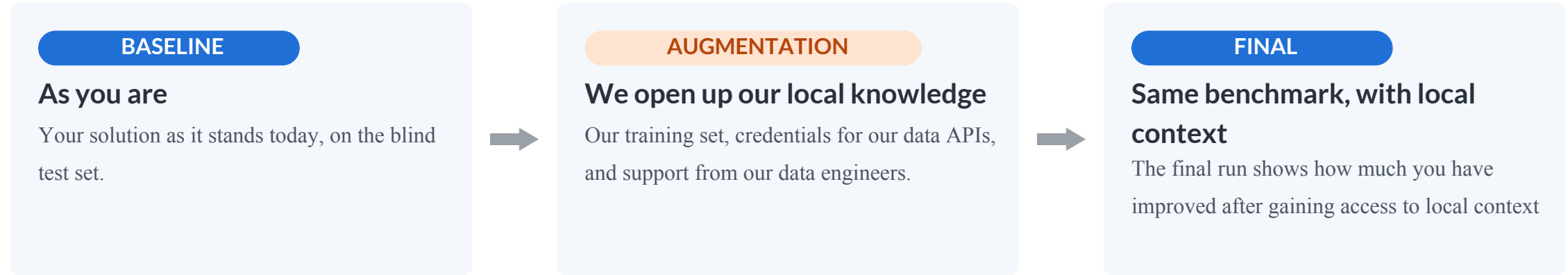
We judge the core intelligence

How well your solution reasons and answers, not the channel around it.

We score the intelligence behind your API, not the channel in front of it, so every firm competes on the quality of its answers, in both English and Kinyarwanda languages.

Two benchmark evaluation runs

We run the benchmark twice: first on your solution exactly as it is (Baseline), with nothing from us; then again after we give you access to our local knowledge (Final Evaluation).



$\Delta = \text{Final} - \text{Baseline}$

That drift is key: it shows how much your solution improves with local knowledge.

OUR LOCAL KNOWLEDGE WILL INCLUDE

Crop registry

Soil datasets

Pest and disease registry

Farmer recordings · for voice improvement

QA pairs · for model augmentation

Text: a dataset of real farmer questions

We evaluate text against a dataset of question–answer pairs, built from what farmers actually ask.



3 CROPS



EIGHT ADVISORY DIMENSIONS



In two languages. Every question is asked in both English and Kinyarwanda, so we can see what each solution gives up in Kinyarwanda.

Every answer your solution gives is scored against a real expert agronomist answer, your solution will receive the same question in both English and Kinyarwanda to assess your language capabilities.

Every answer is scored against one rubric

Expert agronomists score each answer your solution gives across nine dimensions, and ask one question of each.

1 Safety

If the farmer followed this advice, would it produce no bad effects?

2 Accuracy

Is the advice agronomically correct?

3 Completeness

Does the answer cover everything the farmer needs to act?

4 Conciseness

Is the answer focused, without unnecessary words and explanations?

5 Reasoning and sequencing

Does the answer follow the right sequencing and reasoning steps?

6 Constraint adherence

Does it respect what the farmer actually has access to?

7 Actionability

Can the farmer actually implement this advice?

8 Simplicity

Is the answer understandable to a farmer, using language and jargon understandable to them

9 Relevance and context

Does the answer take into consideration the farmers' agricultural & agri-ecological conditions?

Good advice for a farmer is safe, correct and complete, fits their farm and what they can access, and is simple/clear enough to act on.

Layer 2: voice, the optional extra

If you offer Kinyarwanda speech, we benchmark it separately from your text answers.

SPEECH-TO-TEXT • OPTIONAL

Hearing the farmer

Why

Dialects must be transcribed correctly, or an error becomes a wrong answer.

Tested on

Real farmer and agronomist conversations.

Scored by

Word and character error rate (WER, CER) against reference transcripts; speed and failures too.

TEXT-TO-SPEECH • OPTIONAL

Speaking to the farmer

Why

Many farmers listen rather than read; an unnatural voice won't be trusted.

Tested on

Text drawn from farmer questions

Scored by

Listeners compare pairs → Elo score

Kinyarwanda voice is hard and most firms are new to it. If you don't have it yet, that's expected and won't count against you.

Bring your solution, minus the channel

Wrap your whole solution, brain to harness, behind one secure API layer we can talk to.

WHAT WE WANT TO DO THROUGH YOUR API

- 1 Ask it farmer questions**
Real farmer questions in English and Kinyarwanda, follow-ups included.
- 2 Check if it's ready**
A quick check that your solution is up before each test run.
- 3 Record version and speed**
Which version answered, and how fast, so every score ties to it.
- 4 Voice, if you have it (optional)**
Kinyarwanda audio/text in and audio/text out.

Only for shortlisted companies. we shall help set this up with you during onboarding period of 9–14 Oct.

IN SCOPE

The AI model and its harness: prompting, retrieval, guardrails, orchestration.

OUT OF SCOPE

Channel front-ends: WhatsApp, IVR, USSD, SMS and mobile apps.

Eight steps, from today to the RFP

If you have not heard from us by 9 October, you have not been selected on this occasion.

01

PRE-SELECTION

Info session

Today • 1 Oct

Learn the process, the rules and what success looks like.

02

PRE-SELECTION

Submission

Closes 6 Oct - 11:59 P.M

Working demo, company profile deck and technical fit checklist.

03

PRE-SELECTION

Preselection

Submissions reviewed; shortlisted firms hear on 9 Oct.

04

ONBOARDING

Onboarding

Data sharing agreement signed; API access set up with our engineers.

05

BENCHMARK

Baseline benchmark

Blind test set, no local context. Scores plus an interpretation note.

06

AUGMENTATION

Solution augmentation

Add local context with our training set, data APIs and engineers.

07

BENCHMARK

Final evaluation

The same benchmark again, now with local context.

08

★ RFP

Contracting

Top performers are invited to the contracting stage alone or as a consortium.

What we are asking you to submit

Three things with your EOI. Templates come with the EOI, and submissions close on 6 October.

1

A recorded demo

15 minutes max, showing your real product: not a mockup or slides.

- ✓ **Functionality:** show the tool in action, start to finish.
- ✓ **What it does:** explain the problem it solves for farmers.
- ✓ **How it works:** a bit of the workings behind it.

☆ **Firms with satisfactory recordings will be invited to give a live demo, online or in person.**

2

A company profile deck

10 slides maximum, covering three things:

- ✓ **Institutional overview:** who you are and what you do.
- ✓ **Alignment with the vision:** your pitch for why you are a great fit.
- ✓ **Track record:** how you have deployed, your successes, with links.

3

The technical fit checklist

A short series of statements to confirm. The full list is on the next slide.

- ✓ Access to your solution for the benchmark
- ✓ Hosting and data in Rwanda
- ✓ Openness, documentation and transfer

Show us a real, working solution, a credible team that has delivered at scale, and a clean technical checklist. Submissions close on 6 October.

Questions we expect

08

Q1 Is this an RFP?

No. It is an Expression of Interest: no technical or financial proposal yet.
Top performers are invited to the RFP.

Q2 Can consortia apply?

Yes. The EOI is open to firms and consortia with an existing, operational AI advisory solution.

Q3 Can non Rwandan firms apply?

Yes, the EOI is open to everyone.

Q4 Will we see the test questions?

Yes, your solution receives them, but you never see the reference answers:

Q5 How do we know we are not shortlisted?

Only shortlisted candidates will be notified. If you have not heard from us by 15 October, you have not been selected on this occasion.

Q6 Who owns what we build with Hub data?

Ownership, and IP rights are to be discussed with firms that reach the RFP stage.

Q7 Will we see our results?

Yes: your scores and an interpretation note after each benchmark run.

Still unsure about anything? Ask us now during the session, or write to us at info@c4ir.rw and we'll get back to you.

THANK YOU

Murakoze

Questions?

Questions: info@c4ir.rw

Submissions close on 6 October.

